CHAPTER 14

The Actinide and Transactinide Elements

Contents

14.1. Early "Transuranium" Elements 406
14.2. The Actinide Elements 410
14.2.1. Actinium, element 89 410
14.2.1.1. Isotopes *410*
14.2.1.2. Occurrence and production *410*
14.2.1.3. Uses *410*
14.2.2. Thorium, element 90 410
14.2.2.1. Isotopes *410*
14.2.2.2. Occurrence and production *411*
14.2.2.3. Uses *411*
14.2.3. Protactinium, element 91 412
14.2.3.1. Isotopes *412*
14.2.3.2. Occurrence and production *412*
14.2.3.3. Uses *413*
14.2.4. Uranium, element 92 413
14.2.4.1. Isotopes *413*
14.2.4.2. Occurrence, resources and production capacity *413*
14.2.4.3. Production techniques *415*
14.2.4.4. Production wastes *417*
14.2.4.5. Uses *418*
14.2.5. Neptunium, element 93 418
14.2.5.1. Isotopes *418*
14.2.5.2. Occurrence and production *419*
14.2.5.3. Uses *419*
14.2.6. Plutonium, element 94 419
14.2.6.1. Isotopes *419*
14.2.6.2. Occurrence and production *419*
14.2.6.3. Uses *420*
14.2.7. Americium and curium, elements 95 and 96 420
14.2.7.1. Isotopes *422*
14.2.7.2. Occurrence and production *422*
14.2.7.3. Uses *423*
14.2.8. Berkelium and californium, elements 97 and 98 423
14.2.8.1. Isotopes *423*
14.2.8.2. Occurrence and production *423*
14.2.8.3. Uses *424*
14.2.9. Einsteinium and fermium, elements 99 and 100 424
14.2.10. Mendelevium, nobelium and lawrencium, elements 101 – 103 424

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14.2.11. The transactinide elements, $Z \geq 104$ 427
14.2.11.1. Synthesis of elements with $Z \geq 104$ *428*
14.3. Actinide Properties 430
14.3.1. The actinide series 430
14.3.2. The solid state 432
14.3.3. Actinide oxidation states 432
14.3.4. Actinide complexes 433
14.4. Uses of Actinides 439
14.5. Transactinide Properties 441
14.5.1. Chemistry of the transactinide elements 442
14.6. The End of the Periodic Table 443
14.7. Exercises 443
14.8. Literature 444

14.1. EARLY "TRANSURANIUM" ELEMENTS

After the discovery of the neutron by Chadwick in 1932, the group led by Fermi in Rome began bombarding different elements with neutrons to study the radioactivity induced through (n,γ) reactions. The decay characteristics of the radioactivity in neutron-irradiated uranium was interpreted to indicate that some of the products were probably transuranium elements. Similar conclusions were reached by other research groups; for example in 1937 Meitner, Hahn and Strassman in Berlin proposed the following reaction/decay series (omitting mass numbers):

$$_{92}\mathrm{U} + \mathrm{n} \rightarrow {}_{92}\mathrm{U} \xrightarrow[10\ \mathrm{s}]{\beta} {}_{93}\mathrm{EkaRe} \xrightarrow[2.2\ \mathrm{m}]{\beta} {}_{94}\mathrm{EkaOs} \xrightarrow[59\ \mathrm{m}]{\beta} \text{etc} \qquad (14.1)$$

However, many anomalies existed in the chemical properties of these "transuranium elements" as pointed out by Ida Noddack, Irene Curie, Savitch and others.

In 1939 Hahn and Strassman conducted a series of extremely careful chemical investigations which showed that these "transuranium elements" were in fact isotopes of elements in the middle of the Periodic Table, such as Sr, La, Ba etc. The suggestion that their observations were due to a fission of uranium was first made in a scientific paper by Ida Noddack, but ignored. About two years later Meitner and Frisch postulated that the uranium atoms were caused to split by thermal neutrons into two approximately equal fragments.

Further investigation showed that it was the isotope $^{235}\mathrm{U}$ which had undergone fission, and that large amounts of energy and about 2.5 neutrons were released in the process. It occurred to a number of scientists that, if at least one of these neutrons could be captured by other uranium atoms to cause further fission, a chain reaction should be the consequence. This led to the development of nuclear energy and nuclear weapons, as described in Chapter 19.

To the present time 21 transuranium elements have been synthesized; the isotopes known are summarized in Figure 14.1; except elements above 114, which are still uncertain.